

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0572

Neely 138 kV
April 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0572. The Interconnection Customer proposes a 44.2 MW generating facility interconnecting at the Neely 138 kV substation in the Virginia area. The request was received on January 12, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 279 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$336,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Reach Substation-Candelaria 138 KV	Bear-Clover Substation 345 KV	133.8	293	5.1
Monroe Ne-Cedar Crk 345 KV	Monre Ne-Co Line Road 500 KV	109.4	1,144	6.2
Banning-Desert Outpost 138 KV line	South Texa-Franklin-Elk Grove 138 KV line	100.5	190	5.8
Vc Porterdale Hydroelectric-Bonsack 138 KV	Lone Tre-Craven County Wood Energy Lp 138 KV	100.5	269	44.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Reach Substation-Candelaria 138 KV	Rebuild line with higher-capacity conductor	\$9,999,000	1% (\$136,000)
Monroe Ne-Cedar Crk 345 KV	Reconductor line	\$8,888,000	1% (\$57,000)
Banning-Desert Outpost 138 KV line	Reconductor line	\$5,487,000	1% (\$60,000)
Vc Porterdale Hydroelectric-Bonsack 138 KV	Reconductor line	\$1,036,000	8% (\$83,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$336,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.